PRAKTIKUM BASIS DATA

MODUL 3

TP

Oleh:

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PROGRAM STUDI S1 REKAYASA PERANGKAT LUNAK

DIREKTORAT KAMPUS PURWOKERTO

UNIVERSITAS TELKOM

2026

TP

1. Tabel

Tabel Mahasiswa

Worksheet | Query Builder

```
CREATE TABLE mahasiswa (
    id NUMBER PRIMARY KEY,
    nama VARCHAR2(50),
    tempat_lahir VARCHAR2(50),
    tanggal_lahir DATE,
    nomor_hp VARCHAR2(15),
    email VARCHAR2(50),
    tinggi_badan NUMBER,
    berat_badan NUMBER
);

INSERT INTO mahasiswa VALUES (1,'Ahmad Fauzi','Bandung',TO_DATE('2002-01-10','YYYY-MM-DD'),'081234567001','ahmad@gmail.com',170,65
INSERT INTO mahasiswa VALUES (2,'Rizky Pratama','Jakarta',TO_DATE('2001-02-11','YYYY-MM-DD'),'081234567002','rizky@gmail.com',168,
INSERT INTO mahasiswa VALUES (3,'Siti Nurhaliza','Surabaya',TO_DATE('2002-03-12','YYYY-MM-DD'),'081234567003','siti@gmail.com',160,
INSERT INTO mahasiswa VALUES (4,'Muhammad Iqbal','Semarang',TO_DATE('2001-04-13','YYYY-MM-DD'),'081234567004','iqbal@gmail.com',17
INSERT INTO mahasiswa VALUES (5,'Dewi Lestari','Yogyakarta',TO_DATE('2002-05-14','YYYY-MM-DD'),'081234567005','dewi@gmail.com',158,
INSERT INTO mahasiswa VALUES (6,'Agus Setiawan','Bandung',TO_DATE('2001-06-15','YYYY-MM-DD'),'081234567006','agus@gmail.com',175,7
INSERT INTO mahasiswa VALUES (7,'Putri Ayuningtyas','Denpasar',TO_DATE('2002-07-16','YYYY-MM-DD'),'081234567007','putri@gmail.com',
INSERT INTO mahasiswa VALUES (8,'Rudi Hartono','Medan',TO_DATE('2001-08-17','YYYY-MM-DD'),'081234567008','rudi@gmail.com',169,64);
INSERT INTO mahasiswa VALUES (9,'Lina Marlina','Padang',TO_DATE('2002-09-18','YYYY-MM-DD'),'081234567009','lina@gmail.com',159,49)
INSERT INTO mahasiswa VALUES (10,'Bayu Saputra','Solo',TO_DATE('2001-10-19','YYYY-MM-DD'),'081234567010','bayu@gmail.com',171,67);
INSERT INTO mahasiswa VALUES (11,'Nabila Zahra','Bandung',TO_DATE('2002-11-20','YYYY-MM-DD'),'081234567011','nabila@gmail.com',157,
INSERT INTO mahasiswa VALUES (12,'Fahmi Ramadhan','Bogor',TO_DATE('2001-12-21','YYYY-MM-DD'),'081234567012','fahmi@gmail.com',173,
```

SQL | All Rows Fetched: 15 in 0.002 seconds

	ID	NAMA	TEMPAT_LAHIR	TANGGAL_LAHIR	NOMOR_HP	EMAIL	TINGGI_BADAN	BERAT_BADAN
1	1	Aditio Nugroho	Semarang	15-03-2004	081234567890	aditio@email.com	170	65
2	2	Budi Santoso	Bandung	21-07-2003	081234567891	budi@email.com	168	60
3	3	Citra Lestari	Jakarta	11-01-2004	081234567892	citra@email.com	160	50
4	4	Dimas Pratama	Surabaya	09-12-2003	081234567893	dimas@email.com	175	70
5	5	Eka Putri	Yogyakarta	02-05-2004	081234567894	eka@email.com	158	48
6	6	Fajar Hidayat	Solo	19-08-2003	081234567895	fajar@email.com	172	68
7	7	Gina Maharani	Medan	23-09-2004	081234567896	gina@email.com	162	52
8	8	Hendra Wijaya	Palembang	30-04-2003	081234567897	hendra@email.com	180	75
9	9	Intan Permata	Makassar	14-02-2004	081234567898	intan@email.com	159	49
10	10	Joko Susilo	Malang	27-06-2003	081234567899	joko@email.com	173	67
11	11	Karin Amelia	Bali	05-10-2004	081234567800	karin@email.com	165	55
12	12	Lukman Hakim	Aceh	18-11-2003	081234567801	lukman@email.com	169	62
13	13	Maya Sari	Padang	07-01-2004	081234567802	maya@email.com	161	51
14	14	Nanda Saputra	Pontianak	29-03-2003	081234567803	nanda@email.com	174	69
15	15	Oki Prasetyo	Batam	12-12-2004	081234567804	oki@email.com	171	66

2. Fungsi Karakter dan string

Worksheet | Query Builder

```
SELECT
LOWER(nama) AS huruf_kecil,
UPPER(nama) AS huruf_besar,
INITCAP(nama) AS huruf_kapital,
CONCAT(nama,email) AS gabungan_string,
LENGTH(nama) AS panjang_nama,
SUBSTR(nama,1,4) AS potong_nama,
LPAD(nama,15,'*') AS lpad_kiri,
RPAD(nama,15,'*') AS rpad_kanan,
LTRIM(nama) AS ltrim_kiri,
RTRIM(nama) AS rtrim_kanan,
INSTR(email,'@') AS posisi,
REPLACE(email,'gmail','student') AS ganti_string
FROM mahasiswa;
```

Query Result | SQL | All Rows Fetched: 15 in 0.005 seconds

	HURUF_KECIL	HURUF_BESAR	HURUF_KAPITAL	GABUNGAN_STRING	PANJANG_NAMA	POTONG_NAMA	LPAD_KIRI	RPAD_KA
1	aditio nugroho	ADITIO NUGROHO	Aditio Nugroho	Aditio Nugrohoaditio@email.com	14	Adit	*Aditio Nugroho	Aditio N
2	budi santoso	BUDI SANTOSO	Budi Santoso	Budi Santosobudi@email.com	12	Budi	***Budi Santoso	Budi San
3	citra lestari	CITRA LESTARI	Citra Lestari	Citra Lestaricitra@email.com	13	Citr	**Citra Lestari	Citra Le
4	dimas pratama	DIMAS PRATAMA	Dimas Pratama	Dimas Pratamadimas@email.com	13	Dima	**Dimas Pratama	Dimas Pr
5	eka putri	EKA PUTRI	Eka Putri	Eka Putrieka@email.com	9	Eka	******Eka Putri	Eka Putr
6	fajar hidayat	FAJAR HIDAYAT	Fajar Hidayat	Fajar Hidayatfajar@email.com	13	Faja	**Fajar Hidayat	Fajar Hi

3.Fungsi Tanggal Dan Waktu

Worksheet | Query Builder

```
SELECT
SYSDATE AS waktu_sekarang,
LAST_DAY(tanggal_lahir) AS akhir_bulan,
NEXT_DAY(tanggal_lahir,'MONDAY') AS hari_berikut,
ADD_MONTHS(tanggal_lahir,6) AS tambah_tgl_bulan,
MONTHS_BETWEEN(SYSDATE,tanggal_lahir) AS selisih_bulan,
ROUND(tanggal_lahir,'YEAR') AS pembulatan_tahun,
TO_CHAR(tanggal_lahir,'MM') AS bulan,
TO_CHAR(tanggal_lahir,'MON') AS bulan_singkat,
TO_CHAR(tanggal_lahir,'MONTH') AS bulan_lengkap,
TO_CHAR(tanggal_lahir,'DD') AS hari,
TO_CHAR(tanggal_lahir,'DY') AS hari_singkat,
TO_CHAR(tanggal_lahir,'YYYY') AS tahun,
TO_CHAR(tanggal_lahir,'YY') AS tahun_pendek,
TO_CHAR(SYSDATE,'HH24') AS jam,
TO_CHAR(SYSDATE,'MI') AS menit,
TO_CHAR(SYSDATE,'SS') AS detik
FROM mahasiswa;
```

Query Result | SQL | All Rows Fetched: 15 in 0.057 seconds

	AKHIR_BULAN	HARI_BERIKUT	TAMBAH_TGL_BULAN	SELISIH_BULAN	PEMBULATAN_TAHUN	BULAN	BULAN_SINGKAT	BULAN_LENGKAP	HARI	HARI_SINGKAT	TAHUN	
1	31-03-2004	22-03-2004	15-09-2004	263,8016382915173237753882915173237753 88	01-01-2004	03	MAR	MARCH	15	MON	2004	04
2	31-07-2003	28-07-2003	21-01-2004	271,6080899044205495818399044205495818 4	01-01-2004	07	JUL	JULY	21	MON	2003	03
3	31-01-2004	12-01-2004	11-07-2004	265,9306705495818399044205495818399044 21	01-01-2004	01	JAN	JANUARY	11	SUN	2004	04

3. Fungsi Konfrensi

```sql
SELECT
TO_NUMBER('12345') AS contoh_number,
TO_CHAR(tanggal_lahir,'DD-MM-YYYY') AS tanggal_char,
TO_CHAR(tinggi_badan) AS tinggi_char,
TO_DATE('2024-05-10','YYYY-MM-DD') AS contoh_date
FROM mahasiswa
WHERE ROWNUM = 1;
```

Query Result

SQL | All Rows Fetched: 1 in 0.022 seconds

	CONTOH_NUMBER	TANGGAL_CHAR	TINGGI_CHAR	CONTOH_DATE
1	12345	15-03-2004	170	10-05-2024

4. Fungsi Numerik

```sql
SELECT
ABS(-10) AS nilai_abs,
MOD(10,3) AS sisa_bagi,
FLOOR(12.7) AS floor_example,
POWER(2,3) AS pangkat,
ROUND(12.567,2) AS pembulatan,
CEIL(12.2) AS ceil_example
FROM dual;
```

Query Result

SQL | All Rows Fetched: 1 in 0.023 seconds

	NILAI_ABS	SISA_BAGI	FLOOR_EXAMPLE	PANGKAT	PEMBULATAN	CEIL_EXAMPLE
1	10	1	12	8	12,57	13